Exercise 1 – Deep Learning (Practical part)
This project implements the LeNet-5 neural network on the Fashion-MNIST dataset and compares the effects of:
1. Baseline (no regularization)
2. Dropout (at the hidden fully-connected layer)
3. Weight Decay (L2 regularization)
4. Batch Normalization

Instructions
The code in colab consists of two cells in the notebook:
Deep_Learning_Ex1.ipynb
To run all models in a single run simply run the first cell, no changes need to be made to the code.
To run a single model only, change the configs variable in the function main() to include just one of the elements currently listed. For example, to run only Dropout, change the configs variable to:
configs = [{"name": "Dropout", "use_dropout": True, "use_batchnorm": False, "weight_decay": 0.0}]

Outputs
The code generates four folders in the directory from which it runs:
1. Plots - contains the 8 convergence graphs requested in the assignment, two convergence graphs per model, one for train and one for test.
2. Plots combined - contains the 4 recommended combined plots, one per model, showing train and test accuracies on a single figure.
3. Checkpoints - contains the final trained model weights:
   - Baseline.pt
   - Dropout.pt
   - WeightDecay.pt
   - BatchNorm.pt
4. runs - TensorBoard logging directory. To use TensorBoard type:
   tensorboard --logdir runs
   in terminal from the directory.

Additionally the code outputs a results.csv file, containing a table of the accuracies for train and test in each configuration.
Attached to the folder the output folders we obtained on our run, that were used in the report.

Load and test weights
To load and test the saved weights, run the last cell in colab after running the first cell. Test accuracies for each model will be printed.